

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EE) Examination

November 2013

EE-414 Power System Operations

Date: 23.11.2013, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Answers the following questions.

[07]

- 1 What is the permissible error for short term load forecasting technique?
- 2 Define the term state estimation.
- 3 Which techniques are used for long term forecasting?
- 4 How mathematically can we say that present data in measurement matrix is bad data?
- 5 Which model has been widely used to represent the behavior of a zero mean stationary stochastic sequence?
- 6 Load forecasting for load time which ranges from few seconds to 5 minutes is required for_____.
(A) State estimation (B) Planning for generation growth (C) Unit commitments
(D) Contingency analysis
- 7 Kalman filtering technique is used for load forecasting since_____.
(A) Time series approach tends to ignore statistical information (B) It does account voltage variation (C) It is useful for state estimation (D) All of the above

Q - 2 (a) Estimate two random variables by least square estimation method for a given [07]
measurement vector z as follows:

$$Z = \begin{bmatrix} 0.50 \\ 0.45 \\ 0.51 \end{bmatrix} \text{ and } H = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

(b) What is the scope of demand side management? Explain the types of DSM measures [07]
and programs.

OR

- Q - 2 (a) In the dc circuit of fig (a) the meter readings are $Z1=9.01A$, $Z2=302A$, $Z3=6.98V$ and $Z4=5.01V$. Assuming that the ammeters are more accurate than the voltmeters, let us assign the measurement weight $W1=100$, $W2=100$, $W3=50$, and $W4=50$, respectively. Determine the weighted least-squares estimates of the voltage sources $V1$ and $V2$. [07]

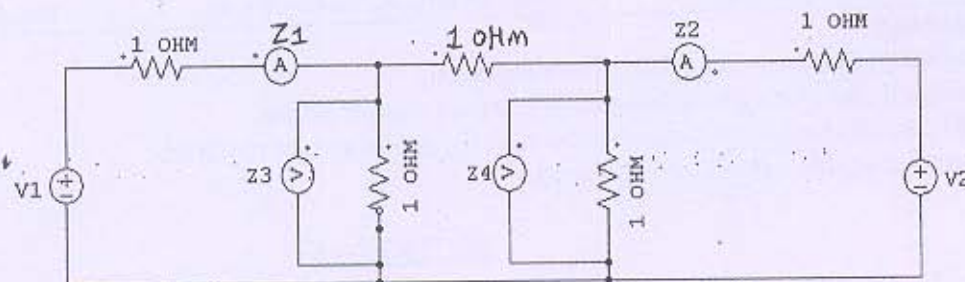


Fig (a)

- (b) Draw the DSM planning stages diagram. Also explain each of the stages in detail. [07]
- Q - 3 (a) Define the term Failure rate and Repair rate. Also derive the expression of probabilities of survival and failure. [07]
- (b) What are the fundamental requirements for communication infrastructure used in distribution automation? Explain any one communication method in detail. [07]

OR

- Q - 3 (a) What is Continuous Markov process? Use continuous markov process to find the probabilities of being available and not being available for a single repairable component having failure rate λ and repair rate μ . [07]
- (b) Write down the short notes on Supervisory Control and Data Acquisition. [07]

SECTION - II

- Q - 4 (a) A system consists of 4 components connected in series having reliabilities of 0.95, 0.92, 0.98 and 0.88. When another system, of 4 similar components in series, is connected in parallel with the system. Find the reliability of the system. [04]
- (b) Explain the different operating states of power system? [03]
- Q - 5 (a) What is the use of sensitivity factors? Explain Generation shift factors and Line outage distribution factors. [07]
- (b) What are reasons for restructuring? Explain each of them in details. [05]
- (c) Draw the incremental fuel cost curve of generator. Why it is not started from zero? [02]

OR

- Q - 5 (a) Draw and explain the flow charts of AC power flow method for contingency analysis. [07]

- (b) Justify the production cost being considered as a function of real-power generation. [02]
- (c) What are the different entities in deregulated environment? [05]
- Q - 6 (a) Explain the structure of traditional vertically integrated electric industry. Also write down the characteristics of traditional vertically integrated electric industry. [07]
- (b) A two bus system is shown in fig (b). If 200 MW is transmitted from plant 2 to the load, a loss of 50MW is incurred. System incremental cost is Rs. 30/ MWh. Find P_{G1} , P_{G2} and power received by load if [07]

$$\frac{\partial F_1}{\partial P_{G1}} = 16 + 0.02P_{G1}$$

$$\frac{\partial F_2}{\partial P_{G2}} = 20 + 0.04P_{G2}$$

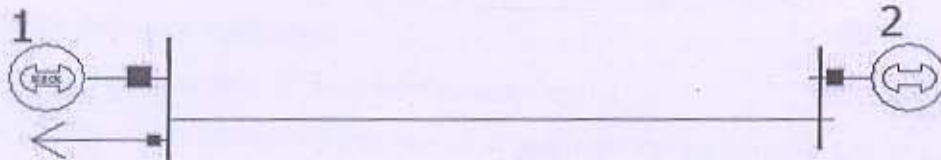


Fig (b)

OR

- Q - 6 (a) Write down the short notes on Congestion management. [07]
- (b) The fuel cost in \$ / h for two 800 MW plants is given by [07]

$$F_1 = 400 + 6.0P_{G1} + 0.004P_{G1}^2 \text{ Rs/h}$$

$$F_2 = 500 + b_2P_{G2} + a_2P_{G2}^2 \text{ Rs/h}$$

Where P_{G1} , P_{G2} are in MW.

- (a) The incremental cost of power is \$8/MWh when total demand is 550MW. Determine optimal generation schedule neglecting losses.
- (b) The incremental cost of power is \$10/MWh when total demand is 1300 MW. Determine optimal generation schedule neglecting losses.
- (c) From (a) and (b) find the coefficients b_2 and a_2 .